

CYLAB SECURITY ACADEMY

Challenge Write-Up Template

Challenge Information

Student Name	Evallo, Daniella Kim N.
Section / Class	BSIT-3G
Challenge Name	Password Profiler
Category	General Skills
Points / Difficulty	Easy
Date Solved	08/10/2026
Flag Format Given	PicoCTF{...}

1. Challenge Description

The challenge provided an SSH connection to a remote instance containing several split file parts. The goal was to combine the parts, identify the resulting file, extract it, and read the flag.

2. Initial Analysis / Reconnaissance

I first reviewed the challenge page and restarted the instance. The instructions provided the SSH host, port, username, and password. After logging in, I checked the home directory and found several files named part_aa through part_ae, indicating that the file had been split into multiple pieces.

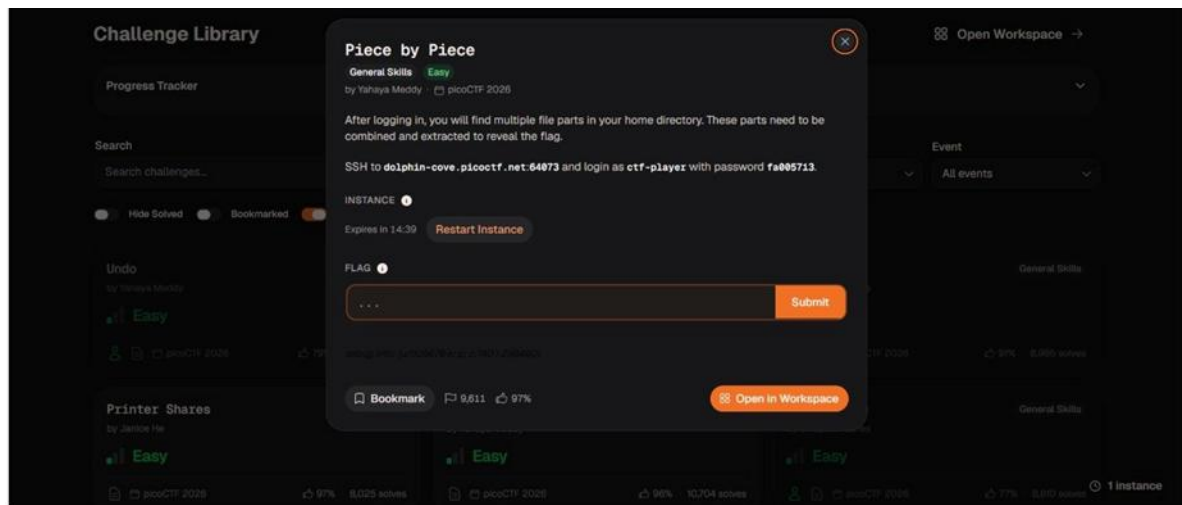
3. Tools and Commands Used

Tool / Command	What it was used for	Result / Output
ssh -p 64073 ctf-player@dolphin-cove.picoctf.net	Connect to the remote challenge instance.	Logged in as ctf-player.
ls -la	List files in the home directory.	Found part_aa through part_ae.
cat part_* > combined	Combine the split file pieces.	Created combined.
file combined	Identify the reconstructed file	Identify the reconstructed file.
unzip combined	Extract the archive.	Extracted flag.txt.
cat flag.txt	Read the extracted flag.	Displayed the picoCTF flag.

4. Solution Walkthrough

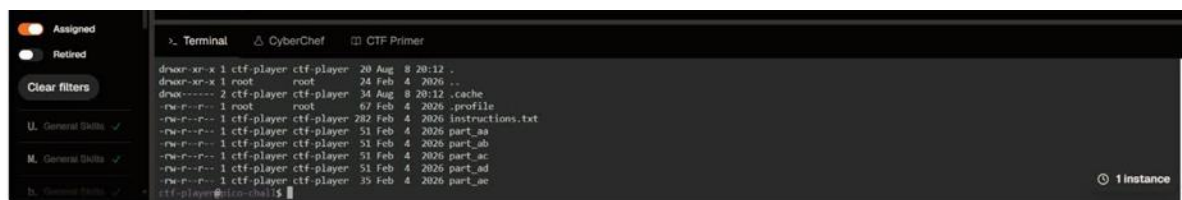
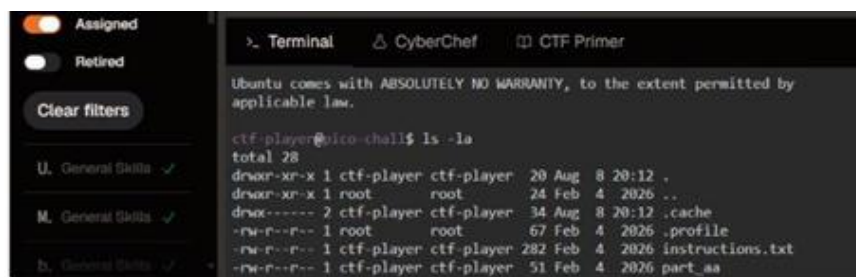
Step 1

I started by reviewing the Piece by Piece challenge instructions and restarting the instance. The challenge provided the SSH host, port, username, and password needed to access the remote Linux environment.



Step 2

I connected to the remote instance through SSH using the provided host, port, username, and password. After logging in, I used `ls -la` to list the files in the home directory.

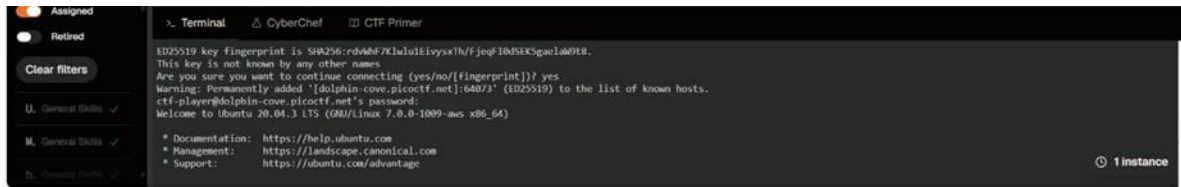


Step 3

I combined the split pieces with `cat part\_\* > combined`. The `cat part\_\*` command reads the matching part files, and `>` redirects their combined contents into a new file named `combined`.

Step 4

This Step shows the SSH command being used to connect: `ssh -p 64073 ctf-player@dolphin-cove.picoctf.net`. The `-p 64073` option specifies the challenge port, while the username and hostname identify the remote instance. The screenshot also shows the host-key confirmation before the connection is established.



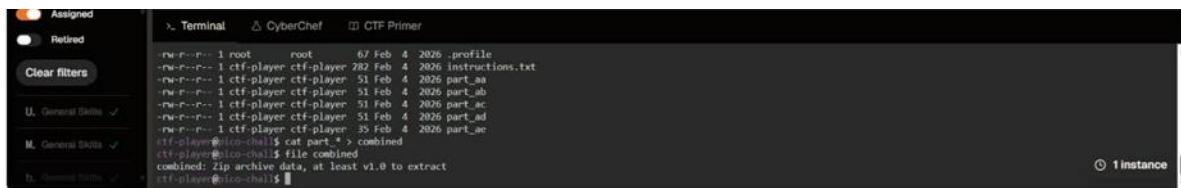
Step 5

I used `file combined` to identify the type of the reconstructed file. The output showed that `combined` was a ZIP archive. I then used `unzip combined` to extract the archive, which produced `flag.txt`.



Step 6

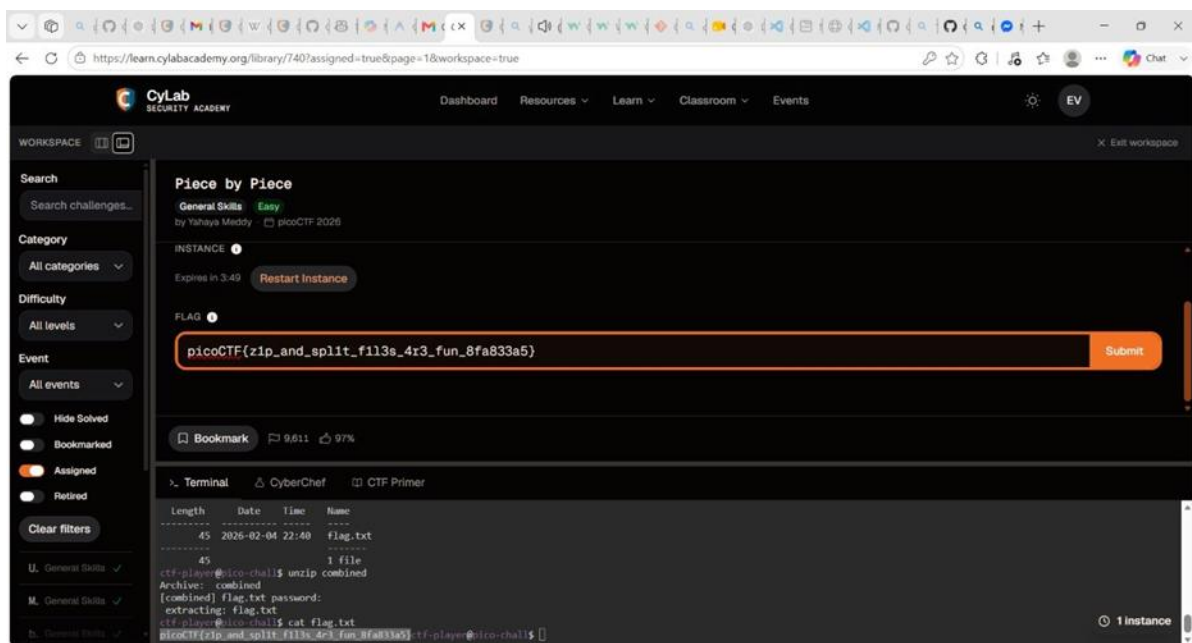
The result of `file combined`: the reconstructed file is recognized as a ZIP archive. This identification is important because it tells me which extraction command to use next.



Step 7

After extracting the archive, I used `cat flag.txt` to display the contents of the extracted file. The terminal showed the picoCTF flag, which I then entered in the challenge submission field.





5. Flag

picoCTF{z1p_and_spl1t_f1l3s_4r3_fun_8fa833a5}

6. Key Concept / What I Learned

This challenge demonstrated file concatenation, archive identification, and extraction in a Linux environment. The important lesson was that split files can be reconstructed with shell commands and then identified and extracted using standard tools.

7. References

YouTube – picoCTF 2026 / CyLab – Piece by Piece

<https://youtu.be/E5ms0r4Z5JE?si=EsJE2D4BdPAdiEtO>